

<https://github.com/fullyloaded/Social-Quantum-Field-Theory/blob/main/README.md>



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Social Quantum Field Theory

Toward a Mathematical Theory of Social Fields

First Edition (2026) Author: Chou I-Hsien

Dedication

To those who believe that mathematics is not merely a collection of formulas, but a language through which the hidden structures of reality may gradually be revealed.

To every mathematician who has sought unity beneath diversity, to every physicist who has transformed geometry into dynamics, to every computer scientist who has discovered structure within information, to every philosopher who has pursued clarity through logic, and to every student who has dared to ask why seemingly unrelated phenomena often obey remarkably similar mathematical principles — this work is respectfully dedicated.

The development of mathematics has always depended upon imagination disciplined by rigor. Every new mathematical language begins with a simple question: can apparently different systems be understood through the same underlying structure? This question inspired analytic geometry, calculus, group theory, differential geometry, Hilbert spaces, information theory, and category theory. It is the same question that motivates the present work.

Social Quantum Field Theory is offered not as the final answer, but as one possible step toward a broader mathematical synthesis. If future generations discover that many of its ideas are incomplete, they should improve them. If some ideas are incorrect, they should replace them. If entirely better formulations emerge, they should be embraced without hesitation. Such is the natural evolution of mathematics.

No mathematical framework belongs permanently to its first author. Every worthwhile theory eventually becomes part of a larger intellectual tradition, refined through criticism, generalization, and collaboration across generations. The greatest hope for this work is

therefore not that it remains unchanged, but that it becomes unnecessary — because future mathematicians will have constructed something even more elegant upon its foundations.

May this book encourage its readers to value rigor over rhetoric, structure over appearance, proof over assertion, curiosity over certainty, and understanding over complexity. For mathematics advances not by defending established ideas, but by continually discovering deeper patterns that unite what once appeared separate.

"The pursuit of structure is the pursuit of understanding."

Dedicated to all researchers who seek unity through mathematics, and all future scholars who will carry that search further.

Publisher's Note

On the Origin and Intended Scope of This Monograph

Social Quantum Field Theory: Toward a Mathematical Theory of Social Fields is presented as a work of mathematical modeling. Its objective is to explore whether concepts drawn from modern mathematics — particularly those inspired by field theory, differential geometry, topology, information theory, and category theory — may provide a unified formal language for the analysis of complex relational systems.

The terminology employed throughout this volume — including expressions such as *field*, *vacuum*, *gauge symmetry*, *entanglement*, *renormalization*, and *excitation* — is used as part of a mathematical framework. Except where explicitly stated, these terms should not be interpreted as asserting that social or organizational phenomena literally obey the laws of quantum mechanics or quantum field theory.

Instead, the work adopts the long-standing mathematical practice of **structural analogy**, in which formal methods developed in one discipline are adapted to another when they preserve essential mathematical relationships. Similar transfers have historically occurred between geometry and mechanics, topology and condensed matter physics, graph theory and computer science, and information theory across numerous scientific domains.

Accordingly, readers are encouraged to distinguish carefully between:

- **Mathematical formalism** — definitions, axioms, theorems, and logical consistency;
- **Computational models** — instantiations of the formalism for specific classes of problems; and
- **Empirical interpretation** — requiring independent validation through observation, experimentation, or statistical analysis.

The present volume primarily addresses the first of these objectives.

Intended Audience

This monograph is intended for readers with interests in mathematical physics, applied mathematics, complex systems, network science, information theory, computational social science, systems engineering, theoretical computer science, and artificial intelligence.

Although examples from social systems occasionally motivate the discussion, the mathematical framework itself is intentionally domain-independent and may be adapted to any sufficiently complex relational system.

A Living Research Program

The author regards this work not as a completed theory but as the initial formulation of an open mathematical research program. Future developments may include refinement of the axiomatic system, stronger analytical results, improved numerical algorithms, additional geometric formulations, operator-algebraic extensions, stochastic and probabilistic generalizations, empirical case studies, and interdisciplinary applications.

Readers are invited to view the framework as an evolving mathematical language rather than a closed or definitive doctrine.

About the Author — Chou I-Hsien

Chou I-Hsien is an independent researcher whose work focuses on the mathematical foundations of complex relational systems. His research interests span applied mathematics, theoretical modeling, information geometry, network science, complex systems, artificial intelligence, and the philosophy of mathematical science.

His work explores whether concepts originating in modern mathematical physics — including variational principles, field theory, differential geometry, topology, operator theory, information theory, tensor networks, and category theory — can be synthesized into a unified mathematical framework for describing large-scale relational phenomena.

The primary outcome of this research program is **Social Quantum Field Theory (SQFT)**, an axiomatic framework that treats relations, rather than isolated entities, as the fundamental mathematical objects of analysis. The framework is intended as a contribution to mathematical modeling and should not be interpreted as asserting that social systems literally obey the laws of quantum mechanics. Instead, it develops structural analogies designed to transfer mathematically rigorous methods across disciplinary boundaries.

Beyond SQFT, Chou I-Hsien is interested in the broader question of how modern mathematics can provide common languages for complex systems that arise in both natural and artificial domains. His work emphasizes the integration of analytical methods, computational techniques, and geometric thinking while maintaining a clear distinction between mathematical formalism and empirical interpretation.

He regards mathematical research as an ongoing collaborative enterprise. Every theoretical framework, in his view, should remain open to refinement, extension, and critical examination. Accordingly, the ideas presented in this volume are intended as the beginning of a long-term research program rather than its conclusion.

Research Interests: Mathematical Foundations of Complex Systems · Social Quantum Field Theory (SQFT) · Variational Principles · Differential Geometry · Topology and Algebraic Topology · Information Geometry · Tensor Network Methods · Network Science · Category Theory · Higher Category Theory · Applied Mathematics · Artificial Intelligence · Computational Modeling · Scientific Computing · Philosophy of Mathematics

Research Philosophy:

"The purpose of mathematics is not merely to describe the world as it appears, but to reveal the structures that remain invariant beneath changing representations."

The SQFT Charter: Foundational Charter

Preamble

The Social Quantum Field Theory (SQFT) Research Community is founded upon the conviction that mathematics advances through open inquiry, rigorous reasoning, transparent criticism, and collaborative refinement.

This Charter does not establish authority over mathematical truth. Rather, it establishes principles for conducting research in a manner consistent with the traditions of modern mathematics.

Membership in this community is defined not by agreement with any particular framework, but by commitment to scholarly integrity.

Article I. Mission

The mission of the SQFT Research Community is:

1. To advance the mathematical study of relational systems;
2. To encourage rigorous proof and transparent reasoning;
3. To develop reproducible computational methods;
4. To compare competing mathematical models fairly;
5. To promote interdisciplinary collaboration where mathematically appropriate;
6. To preserve the historical evolution of the SQFT framework.

Article II. Core Principles

Principle 1 — Intellectual Honesty. All mathematical claims shall accurately distinguish between definitions, propositions, lemmas, theorems, conjectures, computational observations, and empirical hypotheses.

Principle 2 — Openness. Every result is open to verification, correction, refinement, generalization, or replacement by stronger mathematics.

Principle 3 — Transparency. Proofs, assumptions, computational procedures, and datasets should be documented with sufficient clarity to permit independent examination whenever feasible.

Principle 4 — Respect for Prior Work. Research shall acknowledge relevant mathematical literature and distinguish clearly between original contributions and established knowledge.

Principle 5 — Reproducibility. Computational results should, whenever practical, provide algorithms, identify software environments, document parameters, and enable independent reproduction.

Closing Declaration

The SQFT Research Community is founded upon the belief that mathematical progress arises from disciplined curiosity.

Frameworks may change. Notation may change. Proofs may change. Communities may change. The commitment to rigorous inquiry must remain.

"A research community is not defined by unanimous agreement, but by a shared commitment to pursue truth through careful reasoning."

Adopted for the First Edition Archive — Social Quantum Field Theory Research Series Founding Author: Chou I-Hsien Ad Scientiam Communem ("Toward shared knowledge.")

Volume I: Foundations and Formalism

Chapter 1: Introduction (Summary Framework)

The Motivation. Contemporary social, cultural, and technological systems exhibit complex relational dynamics that classical individual-centered models struggle to capture fully. Crises of institutional stagnation, mimetic desire amplification via social media, AI hallucination versus genuine agency, and geopolitical field interactions demand a more powerful formal language.

The Central Question. Can the mathematical structures developed in quantum field theory, information geometry, and open systems provide a rigorous, domain-independent framework for modeling relational social phenomena?

Historical Background. Building upon Bourdieu's field theory (habitus, capital, field as structured space of relations), while addressing remaining gaps in dynamical, nonlocal, and multi-scale description.

Methodological Position. Structural analogy rather than physical reductionism. SQFT employs the formal apparatus of QFT as a mathematical language, not as a claim that society is quantum mechanical.

Contribution. An axiomatic system, operator algebra, geometric and topological tools, and pathways to empirical verification and computational implementation.

Chapter 2: From Bourdieu's Field Theory to Social Quantum Field Theory

Bourdieu's achievement. Fields as structured spaces of objective relations; capital as relational quantity; habitus as internalized structure.

Remaining relational gap. Limited formal treatment of dynamics, nonlocal correlations, structural collapse, and multi-scale evolution.

Transition. From graphs and networks to continuous field ontology. SQFT elevates relations to the fundamental objects, with individuals appearing as localized excitations of the field.

Central Proposition. Social reality is more accurately modeled as an evolving relational field than as an aggregation of independent agents.

Chapter 3: The Mathematical Philosophy of Social Quantum Field Theory

SQFT rests on a deliberate philosophical stance regarding the role of mathematics in the description of complex systems.

Key commitments:

- **Structural Realism over Entity Realism** — What is preserved across different descriptions is the relational structure, not the particular labels of the entities.
- **Analogy with Discipline** — Mathematical structures developed in one domain (quantum field theory, information geometry, topology) may be fruitfully transferred to another domain when the formal relations they encode match the relations of interest. Transfer of structure does not entail transfer of ontology.
- **Openness to Revision** — Every axiom, definition, and theorem is provisional. The framework is offered as a research program rather than a finished doctrine.
- **Distinction of Levels** — Careful separation is maintained between pure mathematical formalism, computational realization, and empirical interpretation and testing.
- **Primacy of Relations** — The fundamental objects of analysis are relations and fields of relations. Individuals and institutions appear as derived, localized excitations or

stable patterns within the field.

This philosophical orientation guides the technical development of subsequent chapters.

Chapter 4: The Eight Axioms of Social Quantum Field Theory

Axiom 1 (Social Hilbert Space). There exists a complex Hilbert space H_S whose rays and density operators represent admissible relational configurations of a social or organizational system.

Axiom 2 (Relational States). The complete description of a relational system at a given time is given by a density operator ρ acting on H_S , satisfying $\rho \geq 0$ and $\text{Tr}(\rho) = 1$.

Axiom 3 (Observables). Physical and social observables correspond to Hermitian operators $\hat{O} = \hat{O}^\dagger$. Expectation values are given by

$$\langle \hat{O} \rangle = \text{Tr}(\rho \hat{O})$$

Axiom 4 (Structural Dynamics). In the absence of environmental coupling, the evolution is unitary and generated by a self-adjoint Structural Hamiltonian $\hat{H}$:

$$i\hbar \frac{d|\Psi\rangle}{dt} = \hat{H}|\Psi\rangle$$

or equivalently the von Neumann equation for density operators.

Axiom 5 (Open Systems). When coupled to an environment, the evolution is described by a completely positive trace-preserving (CPTP) map, typically realized by a Lindblad master equation.

Axiom 6 (Field Operators and Local Excitations). Localized social structures (individuals, organizations, institutions) are represented as excitations created by field operators acting on a relational vacuum.

Axiom 7 (Information Geometry). The space of relational states carries a natural information metric (Fisher–Rao or Bures) that endows it with the structure of a Riemannian manifold. Distances and curvature on this manifold quantify structural difference and stability.

Axiom 8 (Multi-Scale Consistency). Relational descriptions at different scales are related by renormalization-group transformations that preserve the essential dynamical structure while integrating out short-scale degrees of freedom.

These axioms are proposed as a minimal consistent foundation. They are open to refinement as the mathematical and empirical understanding of relational systems advances.

Chapter 5: The Quantum Entanglement Pass — A Worked Model of Relational Coordination

One of the most intuitive illustrations developed in the SQFT series is the *Quantum Entanglement Pass*, using the analogy of a coordinated team sport (e.g., football/soccer).

In the model:

- Players are treated as localized excitations of a shared relational field.
- The successful pass is not merely a classical transfer of an object, but a coordinated correlation of states that can be formalized using entanglement-like measures of mutual information and joint observables.
- The effectiveness of the pass depends on the coherence of the relational field rather than solely on the individual attributes of the players.
- Measurement (observation by opponents or by the environment) can decohere the coordinated state.

This worked example serves both as a pedagogical device and as a concrete testing ground for the operator and information-geometric formalism.

Chapter 6: Social Hilbert Space and the Geometry of Relational Possibility

The Social Hilbert Space H_S encodes the space of admissible relational configurations. Pure states and mixed states (density operators) allow representation of both coherent collective structures and statistical mixtures. The geometry of this space, together with the information metric, supports notions of distance, curvature, and geodesic evolution between relational configurations.

The Social Hilbert Space is not claimed to be a physical Hilbert space of quantum particles. It is a mathematical state space whose linear structure, inner product, and operator algebra provide a powerful language for superpositions of relational configurations, interference of influences, entanglement-like correlations, and open-system dynamics.

The geometry of the space of density operators, equipped with the Bures or quantum Fisher metric, supports a differential-geometric analysis of stability, geodesics between configurations, and curvature as an indicator of structural rigidity or fragility.

Chapter 7: Social Field Operators and Local Excitations

7.1 Field Operators

Definition 7.1 (Social Field Operator). A linear operator $\hat{\Phi}(x, t)$ acting on the Social Hilbert Space H_S , mapping one admissible relational configuration into another:

$$\hat{\Phi}(x, t): H_S \rightarrow H_S$$

Variables x and t represent generalized relational coordinates and temporal evolution.

7.2 Local Excitations

Definition 7.2 (Local Excitation).

$$|\psi_i\rangle = a_i^\dagger |0\rangle$$

where $|0\rangle$ is the relational vacuum state, and $a_i^\dagger$ creates a localized excitation corresponding to an individual, institution, organization, or collective actor.

Definition 7.3 (Relational Vacuum). The vacuum state satisfies $a_i |0\rangle = 0$ for every admissible excitation index, representing the background field prior to localized social structures.

7.3 Creation Operators

Definition 7.4.

$$a_i^\dagger |n_i\rangle = \sqrt{n_i + 1} |n_i + 1\rangle$$

Successive applications generate complex structures:

$$|n\rangle = \frac{(a^\dagger)^n}{\sqrt{n!}} |0\rangle$$

7.4 Annihilation Operators

Definition 7.5.

$$a_i |n_i\rangle = \sqrt{n_i} |n_i - 1\rangle$$

representing disappearance or dissolution of localized structures.

7.5 Number Operator

Definition 7.6.

$$\hat{N}_i = a_i^\dagger a_i, \quad \hat{N}_i |n_i\rangle = n_i |n_i\rangle$$

7.6 Canonical Commutation Relations

$$[a_i, a_j^\dagger] = \delta_{ij}, \quad [a_i, a_j] = 0, \quad [a_i^\dagger, a_j^\dagger] = 0$$

7.7 Field Expansion

$$\hat{\Phi}(x, t) = \sum_i \left(u_i(x, t) a_i + u_i^*(x, t) a_i^\dagger \right)$$

7.8 Interaction Operator

$$\hat{H}_{\text{int}} = \sum_{ij} g_{ij} a_i^\dagger a_j$$

7.13 Structural Hamiltonian

$$\hat{H} = \hat{H}_0 + \hat{H}_{\text{int}} + \hat{H}_{\text{env}}$$

7.14 Open-System Dynamics

$$\frac{d\rho}{dt} = -\frac{i}{\hbar}[\hat{H}, \rho] + \sum_k \left(\hat{L}_k \rho \hat{L}_k^\dagger - \frac{1}{2} \{ \hat{L}_k^\dagger \hat{L}_k, \rho \} \right)$$

Theorem 7.1 (Conservation of Total Probability). If $\text{Tr}(\rho) = 1$ initially, then the Lindblad evolution preserves $\text{Tr}(\rho(t)) = 1$.

Throughout this chapter, the mathematical structures adopted from Quantum Field Theory are employed as formal analogies for relational systems. They should not be interpreted as claims that human society obeys microscopic quantum mechanics.

The chapter establishes the operator-theoretic foundation of Social Quantum Field Theory by adapting the algebraic structure of quantum field operators to relational systems. Together with the Social Hilbert Space introduced in Chapter 6, it provides the mathematical framework required for subsequent developments in topology, gauge structures, renormalization, and empirical applications.

Proposition 7.1. If $g_{ij} = 0$ for all $i \neq j$, then all excitation modes evolve independently.

Proposition 7.2. Localized excitations generated from orthogonal basis functions are linearly independent.

Proposition 7.3. If the interaction Hamiltonian preserves excitation number, then $[\hat{H}, \hat{N}] = 0$, and the total number of active relational excitations remains invariant.

Volume II: Information Geometry, Dynamics and Applications

Chapter 11: Information Geometry and Relational Entropy (Detailed)

Definition 11.1 (Relational Information State). A social field is represented by a density operator (H_S, ρ) .

Definition 11.2 (Relational Entropy).

$$S_R(\rho) = -\text{Tr}(\rho \ln \rho)$$

$$SR(\rho) = -\text{Tr}(\rho \ln \rho)$$

Proposition 11.1. $S_R(\rho) \geq 0$, with equality if ρ is pure. Maximum entropy $\leq \ln N$, achieved when $\rho = I/N$ for an N -dimensional space.

Definition 11.3 (Relative Entropy).

$$D(\rho \parallel \sigma) = \text{Tr}(\rho(\ln \rho - \ln \sigma))$$

Proposition 11.2. $D(\rho \parallel \sigma) \geq 0$, with equality if $\rho = \sigma$.

Definition 11.4 (Mutual Information).

$$I(A:B) = S(A) + S(B) - S(A,B)$$

Fisher Information Metric:

$$g_{ij} = E \left[\left(\frac{\partial \ln p}{\partial \theta^i} \right) \left(\frac{\partial \ln p}{\partial \theta^j} \right) \right]$$

Information Distance: $ds^2 = g_{ij} d\theta^i d\theta^j$

Information Flow: $\partial_\mu J^\mu = \Sigma$, with conservation when $\Sigma = 0$.

Information Potential: $V_I(\rho)$ governs evolution $\frac{d\rho}{dt} = -\nabla_\rho V_I$, subject to stability conditions and defining institutional regions Ω_i within the information manifold.

Entropy Production: $\frac{dS_R}{dt} = \Pi - \Phi$

Theorem 11.1 (Monotonicity of Relative Entropy). Under a CPTP map E ,

$$D(\rho \parallel \sigma) \geq D(E(\rho) \parallel E(\sigma))$$

Proposition 11.4. Institutional resilience increases with mutual information while entropy remains bounded.

Chapter 12: Emergence, Structural Collapse, and Open-System Dynamics (Detailed)

Definition 12.1 (Emergence). Emergence is represented as a collective state $\Phi_E = F(S)$ that is not decomposable into a sum of local excitations: $\Phi_E = F(S) \neq \sum \text{local excitations}$.

Order parameter: $\eta = \langle \hat{O} \rangle$

Structural Potential:

$$V(\eta) = \alpha\eta^2 + \beta\eta^4, \quad \frac{dV}{d\eta} > 0 \text{ (stability condition)}$$

Definition 12.3 (Structural Collapse). A discontinuous transition $\rho_- \rightarrow \rho_+$ via a projection P , occurring at a critical surface $\lambda = \lambda_c$ where a susceptibility $\chi \rightarrow \infty$.

Open-System Evolution (Lindblad):

$$\frac{d\rho}{dt} = -\frac{i}{\hbar} [\hat{H}, \rho] + \sum_k \left(\hat{L}_k \rho \hat{L}_k^\dagger - \frac{1}{2} \{ \hat{L}_k^\dagger \hat{L}_k, \rho \} \right)$$

This evolution includes a dissipation term $D(\rho)$, relaxation toward stationary states ρ^* , memory effects, attractors, basins of attraction, and bifurcations occurring at $\det(J) = 0$.

Resilience Index: $R_s = 1/\tau$

Theorem 12.1 (Stability of Stationary States). If all $\text{Re}(\lambda_i) < 0$ for the Jacobian eigenvalues, then the stationary state is asymptotically stable. Collapse may occur beyond the basin of attraction.

Volume III: Advanced Theory, Control, Inverse Problems and Praxis

Chapter 20: Quantization of Social Fields

20.1 Introduction. The chapter reviews SQFT foundations (relational states, Hamiltonians, open dynamics, renormalization, information geometry, topology) and poses the question: if the social field is a mathematical field, what is its elementary excitation? Individuals are localized excitations of a relational field, not independent units, providing a unified formalism for agency and structure.

20.2 The Relational Vacuum

Definition 20.1 (Relational Vacuum). The reference state $|0\rangle$ with no localized excitations, encoding structural potential (institutions, rules, relationships) without explicit activations:

$$\hat{H}|0\rangle = E_0|0\rangle$$

where $\hat{H}$ is the Hamiltonian and E_0 the ground energy. The vacuum is not empty but potential-laden.

20.3 Local Excitations

Definition 20.2 (Social Excitation). Creation operator $a_i^\dagger$ for relational mode i ; localized excitation $|i\rangle = a_i^\dagger|0\rangle$, representing activated entities (individuals, institutions, etc.).

20.4 Multi-Agent Configurations

$$|\Psi\rangle = a_1^\dagger a_2^\dagger \cdots a_N^\dagger |0\rangle$$

for NN excitations. Configurations are interdependent; ordering matters, unlike classical sets.

Definition 20.3 (Occupation Number). $\hat{N}_i = a_i^\dagger a_i$; $\langle \hat{N}_i \rangle$ measures activation intensity (e.g., participation, communication).

20.5 Commutation Structures

Bosonic: $[a_i, a_j^\dagger] = \delta_{ij}$

These provide algebraic templates for relational constraints (unrestricted vs. exclusive).

20.6 Field Expansion

$$\Phi(x, t) = \sum_k \left(a_k u_k(x, t) + a_k^\dagger u_k^*(x, t) \right)$$

where u_k are mode functions (e.g., regions, sectors).

20.7 Structural Hamiltonian

$$\hat{H} = \sum_k \omega_k a_k^\dagger a_k + \frac{1}{2} \sum_{ijkl} V_{ijkl} a_i^\dagger a_j^\dagger a_k a_l$$

Intrinsic frequencies (first term) and interactions (second term).

Proposition 20.1. If $V_{ijkl} = 0$, modes evolve independently.

20.8 Interpretation. Quantization unifies individuals and society as field excitations; sets stage for interactions in Chapter 21.

Chapter 21: Interaction Vertices and Perturbation Theory of Social Fields

21.1 Introduction. Extends quantization to interactions via vertices representing relational events (information, influence transfer).

21.2 Free and Interaction Hamiltonians

$$\hat{H} = \hat{H}_0 + \hat{H}_{\text{int}}, \quad \hat{H}_0 = \sum_i \omega_i a_i^\dagger a_i, \quad \hat{H}_{\text{int}} = \sum_{ijkl} g_{ijkl} a_i^\dagger a_j^\dagger a_k a_l$$

,

Definition 21.1 (Relational Coupling Constant). g_{ijkl} quantifies influence strength (e.g., communication efficiency).

21.3–21.9 Perturbative expansion, Dyson series, interaction vertices, scattering interpretation, effective interactions, weak and strong coupling.

Theorem 21.1 (Perturbative Validity). If $\|\hat{H}_{\text{int}}\| < \|\hat{H}_0\|$, the perturbative series converges.

Corollary 21.1. Weak coupling allows low-order approximations.

Vertices bridge local to collective dynamics; this leads to strong-coupling phenomena in Chapter 22.

Chapter 22: Spontaneous Symmetry Breaking and Emergent Institutional Order

22.1 Introduction. Describes order emergence without controllers (e.g., consensus, standards) via spontaneous symmetry breaking (SSB).

22.2 Symmetric Relational States

$$[\hat{H}, \hat{U}(g)] = 0 \quad \text{for symmetry group } G$$

Definition 22.1 (Structural Symmetry). Invariance under relational transformations.

22.3 Degenerate Ground States

$$V(\phi) = -\mu|\phi|^2 + \lambda|\phi|^4$$

with minimum at $|\phi| = \frac{\mu}{\sqrt{2\lambda}}$, forming a family of minima M_0 .

22.4 Institutional Ordering. Evolution toward one minimum $\phi = \phi_0$, breaking the symmetry.

Definition 22.2 (Order Parameter). $\Phi = \langle \hat{O} \rangle$: zero in the disordered phase, nonzero in the ordered phase (e.g., consensus strength).

22.5–22.10 Collective modes, Goldstone-type modes, explicit symmetry breaking, structural phase diagram, criticality.

Theorem 22.1 (Emergent Institutional Order). Symmetry + degenerate minima + evolution to one minimum $\Rightarrow$ SSB.

Corollary 22.1. Institutions can be understood as stable ordered phases.

SSB explains decentralized order and leads to the discussion of criticality in Chapter 23.

Chapter 31: Optimal Control and Decision Theory for Relational Fields

This chapter extends SQFT to controlled relational systems, treating interventions like policies or resource allocations as control operators acting on an evolving relational field.

The controlled Hamiltonian becomes:

$$\hat{H}(t) = \hat{H}_0 + \hat{H}_{\text{int}} + \hat{H}_c(u(t))$$

where $\hat{H}_0$ represents intrinsic relational dynamics, $\hat{H}_{\text{int}}$ represents endogenous interactions, and $\hat{H}_c$ represents external interventions.

For open systems, the evolution follows a controlled Lindblad equation:

$$\frac{d\rho}{dt} = -\frac{i}{\hbar}[\hat{H}(u), \rho] + \sum_k \left(\hat{L}_k \rho \hat{L}_k^\dagger - \frac{1}{2} \{ \hat{L}_k^\dagger \hat{L}_k, \rho \} \right)$$

The objective is to minimize a functional:

$$J(u) = \Phi(\rho(T)) + \int_0^T C(\rho, u) dt$$

subject to constraints such as $u_{\min} \leq u(t) \leq u_{\max}$.

Methods include feedback control $u(t) = K(\rho(t))$, model predictive control (MPC) with rolling horizons, multi-objective optimization via Pareto frontiers, robust control under uncertainty, and ethical constraints ensuring transparency and fairness.

Theorem 31.1. Suppose the admissible control set U is compact, the objective functional $J(u)$ is continuous, and the controlled SQFT evolution admits unique solutions for every admissible control. Then there exists at least one optimal control u^* in U that minimizes the objective functional.

Chapter 32: Inverse Social Field Theory – Reconstructing Hidden Relational Dynamics

An **Inverse Social Field Problem** seeks to infer the latent operators, parameters, and relational states that best explain a given collection of empirical observations under the governing equations of SQFT.

Parameter identification:

$$\hat{\Theta} = \underset{\Theta}{\operatorname{argmin}} \| D - O(\rho(\Theta)) \|^2$$

Bayesian inversion:

$$P(\hat{H}, \Theta, \rho \mid D) \propto P(D \mid \hat{H}, \Theta, \rho)$$

Theorem 32.1. Suppose the SQFT model is correctly specified, the true parameters are identifiable, observational data increase without bound, and the prior assigns positive probability to a neighborhood of the true parameters. Then the Bayesian posterior distribution converges toward the true relational Hamiltonian and parameter set.

Chapter 33: Variational Principles and the Principle of Least Action for Relational Fields

The Relational Action:

$$S[\Phi] = \int_{\Omega} L(\Phi, \partial_{\mu}\Phi) d^4x$$

where L is the relational Lagrangian density. The central postulate is $\delta S = 0$.

Applying the calculus of variations yields the **Euler–Lagrange equations**:

$$\frac{\partial L}{\partial \Phi} - \partial_{\mu} \left(\frac{\partial L}{\partial (\partial_{\mu} \Phi)} \right) = 0$$

Theorem 33.1. Suppose the relational action is differentiable, admissible variations vanish on the boundary, and the Legendre transformation is regular. Then every stationary point of the action satisfies the Euler–Lagrange equations, and these equations are locally equivalent to the Hamiltonian equations.

Chapter 34: Noether's Theorem and Conservation Laws in Relational Fields

Associated with every continuous symmetry is a conserved current. Symmetries of the action imply conserved quantities via Noether's theorem. In open systems there may be sources:

$$\partial_{\mu} J^{\mu} = \Sigma$$

Chapters 41–48 Highlights (Advanced Mathematical Structures)

Chapter 41: Quantum Anomalies and Emergent Symmetry Breaking. Anomalies describe violations of classical conservation laws after quantization or coarse-graining. The functional measure may not be invariant, resulting in $\partial_{\mu} J^{\mu} = A$ (an anomaly term).

Chapter 42: Effective Field Theory and Multiscale Relational Dynamics.

$$L_{\text{eff}} = L_0 + \sum_i \frac{c_i}{\Lambda^n} O_i$$

Running couplings and universality emerge through coarse-graining.

Chapter 43: Information-Theoretic Foundations. Shannon entropy, relative entropy, mutual information, information flow, and monotonicity under maps.

Chapter 44: Computational Complexity and Tensor Network Representations. Matrix Product States (MPS), Tree Tensor Networks, PEPS, and MERA for efficient representation of high-dimensional relational states, with polynomial complexity under an area-law entanglement assumption.

Chapter 45: Category Theory and Functorial Structures. Relational categories, functors, natural transformations, monoidal categories, and higher categories.

Chapter 46: Topos Theory and Internal Logic. Topoi with internal logic (Heyting algebra), presheaves and sheaves for local-to-global consistency.

Chapter 47: Homotopy Type Theory. Identity as paths, higher groupoids, and the univalence axiom.

Chapter 48: Derived Geometry. Chain complexes, cohomology, and derived categories for enriched geometric structures.

Praxis: Selected Themes and Conceptual Bridges

Personal Cultivation in Field Terms

Personal cultivation within SQFT is the practice of consciously shaping one's local excitations and selective couplings. Core practices mapped to the formalism:

- **Topological decoupling** — reducing unwanted correlations with stagnant or destructive sub-fields.
- **Observation and the Zeno effect** — repeated measurement (reflection, journaling, public commitment) can suppress unwanted trajectories.
- **Responsibility as source term** — accepting to act as a stable boundary condition or driving term for the surrounding field.
- **Proactive outreach and service declaration** — generating new productive modes and testing the response of the larger field.

The aim is coherent participation rather than isolation.

Institutional Stability and Fragility

Institutions appear as long-lived collective modes or deep effective potentials:

- Emergence through spontaneous symmetry breaking and order-parameter condensation.
- Rigidity associated with high curvature or deep wells in the effective potential.
- Fragility when the basin of attraction shrinks or when external perturbations drive the system across a critical surface.
- Design as the engineering of Hamiltonians, dissipation channels, and topological protection.

Human Roles in Hybrid Human–AI Fields

In an AI-rich environment, distinctive human functions emphasize:

- Final accountability and responsibility-bearing.
- Transmission of non-propositional cultural and emotional knowledge.
- Generation of genuinely novel relational possibilities.
- Ethical judgment under deep uncertainty.

AI systems function as powerful amplifiers and simulators; the anchoring of meaning and liability remains human or clearly designated hybrid.

Market and Asset Fields

Economic systems can be viewed as multi-scale relational fields in which prices, expectations, and capital flows act as order parameters or collective excitations. Possible directions within SQFT include:

- Asset prices as expectation values of certain field operators.
- Liquidity and risk as geometric or topological properties of the configuration space.
- Regime shifts and crashes as structural phase transitions or basin escapes.
- Hard assets as relatively stable modes under high-entropy monetary environments.
- Inflation and stagflation as changes in the effective potential and dissipation structure.

These remain conceptual sketches requiring concrete model building and data calibration.

Taiwan / Technology Ecosystem Sketch

The semiconductor and advanced manufacturing ecosystem centered on Taiwan can be analyzed as a highly coherent, high-value relational field characterized by:

- Strong local couplings and specialized knowledge excitations.
- Global entanglement through supply chains and geopolitical constraints.
- High topological protection arising from accumulated process knowledge and trust networks.
- Sensitivity to external field perturbations (policy, conflict, demand shocks).

SQFT offers a language for discussing resilience, criticality, and multi-scale coordination in such systems without reducing them to simple network graphs or agent-based models alone.

Appendices (Selected)

Appendix G: Future Research Agenda and Open Problems in Social Quantum Field Theory

Open Problem G.1. Determine sufficient conditions for the global existence and uniqueness of solutions to the SQFT master equation

$$\frac{d\rho}{dt} = -\frac{i}{\hbar}[\hat{H}, \rho] + D(\rho)$$

under general nonlinear interaction Hamiltonians.

Open Problem G.2. Construct a rigorous functional-analytic foundation for infinite-dimensional Social Hilbert Spaces, $H \rightarrow H_\infty$. Such an extension would permit the study of large-scale societal systems approaching the thermodynamic limit.

Open Problem G.3. Characterize the spectrum $\sigma(\hat{H}) = \{\lambda_i\}$ and determine how eigenvalue distributions influence institutional stability and structural phase transitions.

Open Problem G.4. Define curvature invariants capable of predicting impending structural collapse before observable crises emerge. One possible direction is to investigate scalar quantities of the form $K = f(g_{ij}, R_{ij}, \nabla R)$.

Open Problem G.5. Develop topological invariants capable of distinguishing institutional persistence, organizational resilience, and irreversible structural transformation. Persistent homology and topological data analysis offer promising mathematical tools.

Open Problem G.6. Develop scalable algorithms whose computational complexity satisfies $O(N \log N)$ or better while preserving positivity and normalization.

Conjecture G.1 (Universality of Relational Dynamics). There exists a finite collection of invariant mathematical structures governing the evolution of all sufficiently complex relational systems, independent of their empirical domain.

Appendix X: Notation Index (Selected)

Symbol	Meaning
H	Hilbert space / Hamiltonian
ρ	Density operator
$\hat{L}_k$	Lindblad operator
S	Entropy
G	Symmetry group
$[A, B]$	Commutator

Symbol	Meaning
--------	---------

$\{A, B\}$	Anticommutator
------------	----------------

Frequently Used Equations: Lindblad Master Equation; Path Integral; Renormalization Group Flow, $\beta(g) = \mu \frac{dg}{d\mu} \beta(g) = \mu d\mu dg$.

Appendix Y: List of Acronyms

- **SQFT** – Social Quantum Field Theory
- **QFT** – Quantum Field Theory
- **RG** – Renormalization Group
- **EFT** – Effective Field Theory
- **MPS** – Matrix Product State
- **PEPS** – Projected Entangled Pair States
- **MERA** – Multi-scale Entanglement Renormalization Ansatz
- **CPTP** – Completely Positive Trace-Preserving
- **OQS** – Open Quantum Systems

Appendices L–W (Summary of Contents)

- **Appendix L:** Mathematical Notation and Symbol Index
- **Appendix M:** Mathematical Dictionary of SQFT
- **Appendix N:** Correspondence between QFT and SQFT (Theorem N.1 on structural mapping)
- **Appendix O:** Comparison with existing theories (network science, dynamical systems, control theory, information theory, etc.)
- **Appendix P:** Relationship to the Philosophy of Science (structural realism)
- **Appendix Q:** Mathematical Roadmap for Future Development
- **Appendix R:** Open Mathematical Problems
- **Appendix S:** Research Agenda and Experimental Validation Framework
- **Appendix T:** Foundational Mathematical Conjectures
- **Appendix U:** Mathematical Glossary of Fundamental Definitions
- **Appendix V:** Mathematical Consistency, Scope, and Limitations
- **Appendix W:** Epilogue – Toward a Unified Mathematical Language of Relational Systems

Principle W.1: Structural priority – relations over isolated objects.

Preliminary Glossary of Key Terms

- **Social Hilbert Space** – The mathematical state space whose elements (or density operators) represent admissible relational configurations.
- **Relational Vacuum** – The structured background state from which localized excitations are created.

- **Local Excitation** — A localized activation of the field corresponding to an individual, group, or institution.
- **Structural Hamiltonian** — The operator generating coherent (non-dissipative) relational dynamics.
- **Lindblad Operator** — Generator of irreversible, open-system effects such as dissipation, decoherence, or environmental monitoring.
- **Order Parameter** — A macroscopic variable distinguishing ordered (e.g., coordinated, institutionalized) from disordered phases.
- **Relational Entropy** — A measure of the uncertainty or mixedness of the relational state.
- **Topological Decoupling** — The controlled reduction or elimination of unwanted correlations with specific sub-fields.
- **Gauge Freedom** — Descriptive redundancy in the choice of coordinates or labels that leaves physical/social content invariant.
- **Information Metric** — A Riemannian metric on the space of relational states (often the Fisher–Rao or Bures metric) quantifying infinitesimal distinguishability.
- **Effective Field Theory (in SQFT)** — A description valid at a given scale, obtained by integrating out shorter-scale degrees of freedom.
- **Spontaneous Symmetry Breaking** — The process by which a symmetric system dynamically selects an ordered configuration that no longer exhibits the original symmetry, giving rise to institutional or collective order.
- **Inverse Social Field Problem** — The reconstruction of latent Hamiltonians, states, or parameters from observed data.
- **Control Operator** — An external intervention term added to the Hamiltonian or Lindblad generators to steer the relational dynamics.
- **Basin of Attraction** — The set of initial states that evolve toward a particular stable configuration or attractor.
- **Structural Collapse** — A discontinuous or rapid transition out of a previously stable relational configuration.

Closing Notes on the Manuscript

On the Use of Quantum Language

A recurring point of clarification is required: the consistent use of terms such as "Hilbert space," "operator," "entanglement," "renormalization," and "vacuum" does not constitute a claim that social systems are quantum mechanical in the physical sense. These terms are employed because they name precise mathematical structures that have proven powerful for describing relational, multi-scale, and open dynamical systems. The analogy is structural and formal, not ontological. Readers are asked to maintain this distinction throughout.

On the Name "Social Quantum Field Theory"

The name is deliberately provocative. It signals both the mathematical sources of the formalism and the domain of intended application. At the same time, it risks misunderstanding. The qualifier "Social" indicates the primary domain of interest; the phrase "Quantum Field Theory" indicates the mathematical toolkit being adapted. The combination is a claim about useful formal analogy, not a claim about the microscopic physics of human beings. Alternative names (*Relational Field Theory*, *Operator Theory of Social Systems*, etc.) remain possible and may eventually prove preferable. For the present, the existing name is retained for continuity with the published materials.

On the Limits of Formalism

No mathematical framework, however elegant, can substitute for domain knowledge, careful measurement, or ethical judgment. SQFT provides a language and a set of structural hypotheses. It does not automatically generate correct models of any particular organization, market, or society. The quality of any concrete application will depend on the skill with which the abstract structures are instantiated and tested.

Invitation to Critique and Extension

This working manuscript is offered in the spirit of open mathematical inquiry. Criticisms of the axioms, alternative formalizations, tighter proofs, counter-examples, and empirical tests are all welcome and necessary for the healthy development of the research program. No framework improves by remaining unchallenged.

Relationship to the Original Blog Series

The present document is a derivative working compilation. The authoritative source for each chapter remains the original posts published on the author's blog in July 2026. In case of discrepancy, the original posts should be consulted. This compilation aims at coherence and cumulative readability rather than verbatim reproduction of every paragraph.

Status of This Working Draft

This document represents a consolidated working manuscript compiled from publicly available First Edition materials released in July 2026, together with structural expansions and clarifications produced during the compilation process. It is not a substitute for the original blog posts, nor is it yet a finished monograph.

What this document currently contains: complete front matter; the Eight Axioms; substantial extracts from Chapters 1–7, 11–12, 20–22, 31–34, and 41–48; summaries of key appendices; conceptual bridges to personal cultivation, institutional analysis, AI-era roles, markets, and geopolitical sketches; open problems and research priorities; and clarifications on the use of quantum language and the limits of formalism.

What remains for a complete First Edition: full text of the remaining Volume I chapters; complete praxis chapters with worked examples; a full set of illustrations and diagrams; a

comprehensive bibliography and index; professional typesetting and bilingual presentation; and systematic empirical illustrations.

The research program remains open, provisional, and dependent on continued mathematical and empirical work.

End of Compiled Manuscript



Toward a Mathematical Theory of Social Fields

Social Quantum Field Theory

and the Formalization of Field Theory

$$i\hbar \frac{\partial |\psi\rangle}{\partial t} = \hat{H} |\psi\rangle$$

$$\frac{d\hat{p}}{dt} = -\frac{i}{\hbar} [H, \hat{p}] + \sum_k (L_k \rho L_k^\dagger - \frac{1}{2} \{L_k^\dagger L_k, \hat{p}\})$$

$$R_{\mu\nu} - \frac{1}{2} g_{\mu\nu} R = 8\pi T_{\mu\nu}$$

$$S = -\text{Tr}(\hat{\rho} \ln \hat{\rho})$$

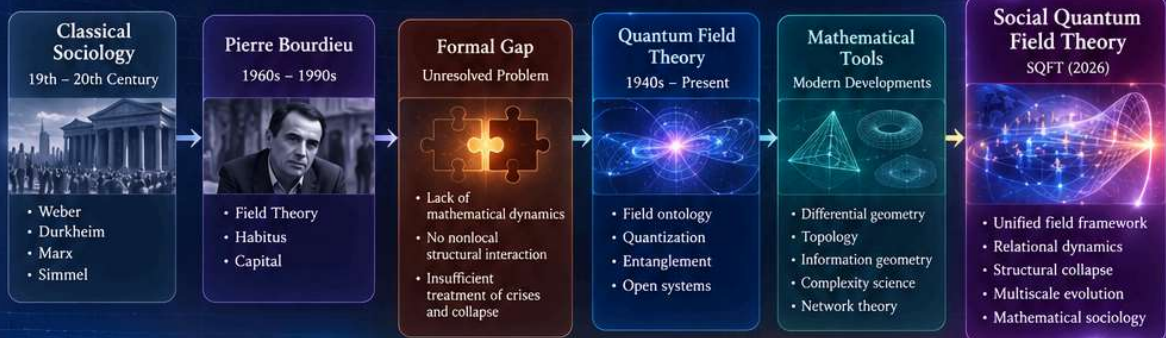
$$Z = \int D\phi e^{iS[\phi]/\hbar}$$

Chou I-Hsien

Preprint Edition

Historical Development of SQFT

From Relational Sociology to a Unified Mathematical Framework



Sociology

Physics

Mathematics

Information

Complexity

AI Era

SQFT

A New Mathematical Language for Understanding Complex Social Systems

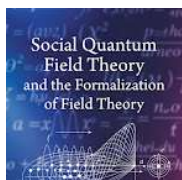


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社會量子場論（SQFT）超越布迪厄的場域理論，建構關係社會學的形式化 1~10

7月 15, 2026



預印本 社會量子場論與場論的形式化 周易先 獨立研究者 · 版本 2.0（數學版） · 2026年7月 摘要 本文提出社會量子場論（SQFT）作為用以表示複雜社會系統中關係動態的形式數學框架。SQFT並不主張社 ...

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量子之影：台灣QNF-3量子導航系統的崛起與其地緣政治影響

9月 27, 2025



量子之影：台灣QNF-3量子導航系統的崛起與其地緣政治影響 摘要 在2035年，全球局勢緊張之際，台灣的量子導航技術成為重塑國際權力格局的關鍵力量。本篇科幻論文式敘事，記錄了國家中山科學研 ...

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